رقم الطالب: _____ اسم الطالب: _____

1. **To transfer value (6F17) 16-BIT data from microprocessor to output port with address 3F0H the following instructions can be used:**

A. MOV DX, 03F0H; MOV BX, 6F17H; OUT DX, BX
B. MOV DX, 03F0H; MOV AX, 6F17H; OUT DX, AX
C. MOV DX, 03F0H; MOV AX, 6F17H; OUT DX, AL
D. MOV DX, 03FCH; OUT DX, AL

2. **The interrupt generated by the execution of an interrupt instruction INT is**

A. NMI interrupt B. software interrupts C. external interrupt D. Interrupt-in-interrupt

3. **The physical address of interrupt vector associated with INT 16 is:**

A. 00042H B. 00040H C. 00064H D. 0003CH

4. **The size of interrupt vectors table is ——**

A. 4 bytes B. 1024 bytes C. 64 bytes D. 128 bytes

5. **The binary value of AL and carry flag (CF)after the following instructions have executed :**

```
MOV AL,0ACH
SAR AL,1
```

A. AL=01010101B; CF=1
B. AL=11010110B;CF=0
C. AL=01010100B; CF=0
D. AL=11010110B;CF=1

6. **The binary value of AL and CF after the following instructions have executed is .....**

```
MOV CL,4
MOV AL, 7FH
ROL AL,CL
```

A. AL=11110111B; CF=0
B. AL=11110111B; CF=1
C. AL=01111111B;CF=1
D. AL=01111111B CF=0

7. **In the following instruction sequence, show the value of AL, in binary:**

```
MOV AL,3FH
NEG AL
XOR AL,FFH
```

A. AL= 00111110B B. AL= 01011111B C. AL= 01011110B D. AL= 01011111B

8. **If there is a carry from lowest 4 bit during addition, _____ flag sets.**

A. Auxiliary carry B. CarryC. Over flow D. Sign

9. **The result of multiplication instruction MUL BX is stored in register pair**

A. DX:AX B. AX:DX C. CX:AX D. AX:CX

10. **Find the content of register AL after execution of the following code:**

```
MOV AL,11
SHR AL,1
ADC AL,1
```

A. 7 B. 5 C. 4 D. 3

11. **Find the content of memory location X after execution of the following code**

```
X DB ?
MOV AX,11
MOV BL,3
DIV BL
OR AH,30H
MOV X,AH
```

A. X='2' B. X=0 C. X= '0' D. X=2

12. **Find the content of AH registers after execution of the following code:**

```
MOV AX,14
MOV BL,4
DIV BL
XOR AH,0FH
```

A. D0H B. C0H C. 0DH D. 0CH

13. **LOPPE instruction repeat a group of instruction while**

A. (CX=0)&(ZF=0) B. (CX=0) & (ZF≠0) C. (CX≠0) &(ZF=1) D. (Cx≠0) & (ZF=0)

Scanned with CamScanner

Student Name:

Part 1: MULTIPLE CHOICES QUESTIONS (20 marks)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20

Part 2: Essay Questions (20 marks)

QUESTION #2

- Write instructions to test bit-5 in register AL, if it is one reset bit-1 otherwise set bit-2

- Write a procedure to store '*' into 1 KB memory locations defined as below:

```
X    DB  1024 DUP(?)
```

- Using BIOS interrupt, write assembler codes to clear screen and set cursor to position (10, 20).

Scanned with CamScanner

- Write an assembler code to reverse string X in order; use LODSB and STOSB strings; where
  X DB 'ABCDEK+++$'

QUESTION#3

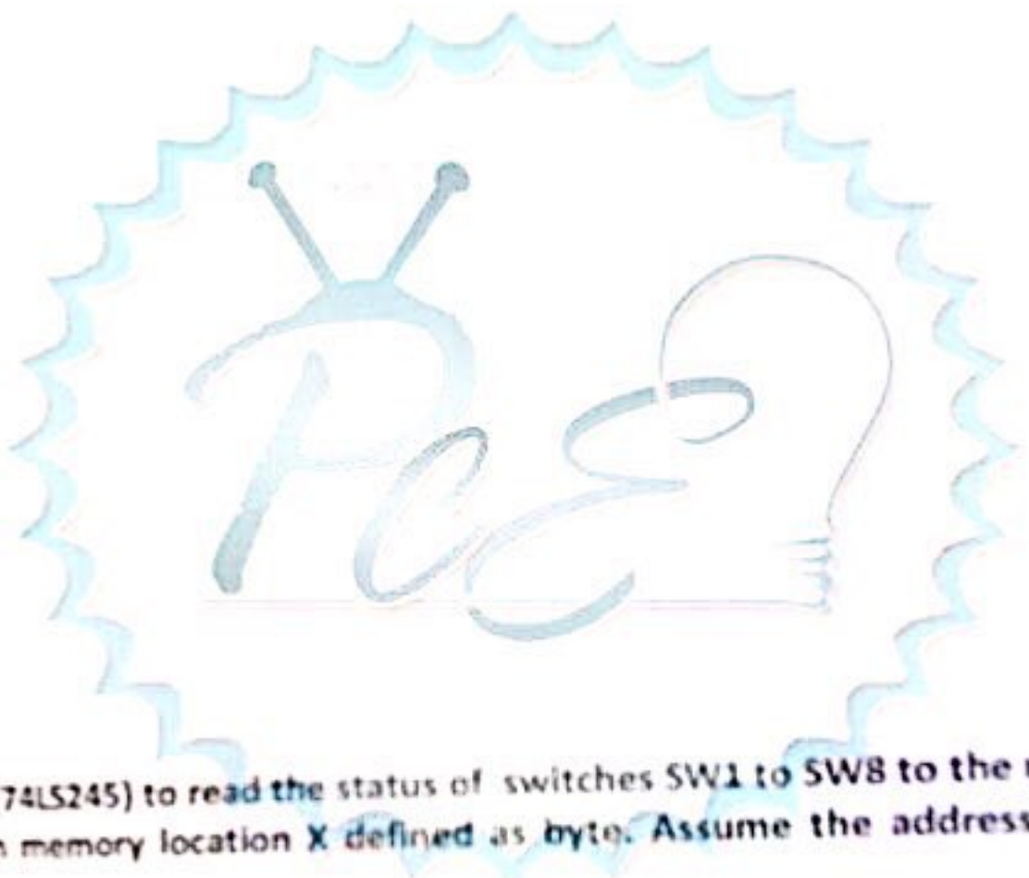
Interface 8- bit input port (74LS245) to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X defined as byte. Assume the address of the port is 08FFH.

a) Draw block circuit diagram circuit, showing address decoding logic, and control signals
b) Write a program to read the status of switches and store the result into X memory location

Scanned with CamScanner

Write an assembler code to reverse string X in order; use LODSB and STOSB strings; where

```
X  DB  'ABCDEK+++*'
```



QUESTION#3

Interface 8- bit input port (74LS245) to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X defined as byte. Assume the address of the port is 08FFH.

a) Draw block circuit diagram circuit, showing address decoding logic, and control signals
b) Write a program to read the status of switches and store the result into X memory location

Scanned with CamScanner